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Assignment-5

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Subject: Statistical Method and Probability theory
Section: I

Q. A1 The purpose of a t-test is to determine if there is a significant difference between the means of two groups.

Q. A2 What is null hypothesis in a t-test?
Sol: the null hypothesis in a t-test states there is no significant difference between the means of two groups
ie $H_0 : \mu_1 = \mu_2$

Q. A3 An F-test is typically used to compare the varIences of two or more groups to determine if they differ significantly.

Q. A4 What does ANOVA (Analysis of Variance) help us test?
Sol: It is a statistical method to analyze differences among group means in a sample

Q A5 In a paired samples t-test, each observations in one group is Matched with an observation in the other group.

Q A6 What is the null hypothesis in paired-sample t-test?
Sol: $H_0 : \mu_1 = \mu_2$

Q A7 The alternate hypothesis in a paired samples t-test states that the mean difference between paired observation is Not equal to zero.

Q A8 What is the degree of freedom in a paired samples t-test with n number of pairs.

Sol⇒ df ⇒ n-1

Q A9. To Test Independence of attributes, the observed data are arranged in a contingency table.

Q A10. How is the F-ratio calculated in an F-test?

Sol⇒ $$\text{F-ratio} = \frac{\text{Variance between groups}}{\text{Variance } \cancel{\text{between}} \text{ within groups}}$$

Q. B1. The heights of 10 adult males selected at random from a given locality had a mean of 158 cm and variance 39.0625 cm. Test at ~~50~~ 5% level of significance, the hypothesis that the adult males of the given locality are on the average less than 162.5 ~~a~~ cm tall.

Sol⇒ given, $\alpha = 0.05$,

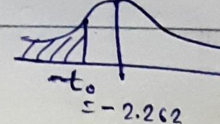
$S = 39.0625$

$\bar{x} = 158$

$n = 10$

(1) Null hypothesis: $H_0: \mu = 162.5$

Alternate: $H_1: \mu < 162.5$ (left tailed)



(ii) $\alpha = 0.05$

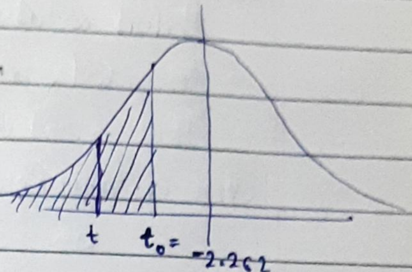
(iii) $df = n-1 = 10-1 = 9$

(iv) critical value →

for $df = 9$, one tailed exp, $\alpha = 0.05$

$$t_0 = -2.262$$

(v) test statistics →



$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$t = \frac{158 - 162.5}{\frac{39.0625}{\sqrt{10}}}$$

$$t = -0.3643$$

Since t lie in critical region, ie we ~~fail~~ reject H_0 hence there is sufficient evidence to say avg height of adult males in given locality is less than 162.5 cm

Q.B2 Two types of batteries, X and Y are tested for their length of life and the following results are obtained

Batteries	Sample Size	Mean (hours)	Variance (hours)
A	10	1000	100
B	12	2000	121

Is there significance difference between the two mean ?

Soln →

(i) $H_0 : \mu_A = \mu_B$

$H_1 : \mu_A \neq \mu_B$ (two tailed exp)

(ii) $\alpha = 0.05$ (standard value)

(iii) $df = n_1 + n_2 - 2$

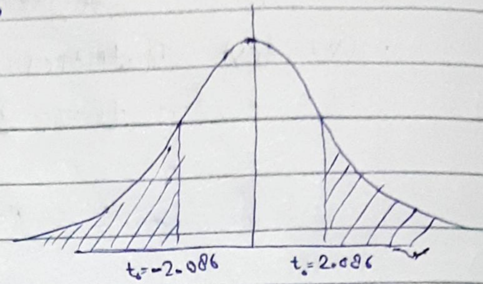
$= 10 + 12 - 2 = 20$

(iv) Critical value :-

for $\alpha = 0.05$ & $dof = 20$

$t_0 = \pm 2.086$

(v) Critical Region :-



(vi) test stastics :-

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_A - \mu_B)}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{---} (1)$$

where $S = \sqrt{\dfrac{n_1 S_A^2 + n_2 S_B^2}{(n_1 + n_2 - 2)}}$

given

$n_1 = 10, \; n_2 = 12$

$\mu_A = \mu_B$ (assumed)

$\bar{x} = 1000, \; S_A^2 = 100$

$\bar{y} = 2000, \; S_B^2 = 121$

$S = \sqrt{\dfrac{10(100) + 12(121)}{(10 + 12 - 2)}}$

$S = 11.0724$

hence test statistics :-

$$t = \frac{(1000 - 2000) - (0)}{11.0724\sqrt{\frac{1}{10} + \frac{1}{12}}}$$

$t = -210.92$

Since test stastics lie in critical region,
ie we reject H_0 ie mean life of X and Y are not equal,
hence there is Significant difference between means. Ans

Q. 83 A travel agency's marketing brochure indicates that the standard deviations of hotel rates for two cities are the same. A random sample of 13 rooms in a city has a standard deviation of \$ 27.50 and a random sample of 16 hotel room rates in the other city has a standard deviation of \$ 29.75. Can you reject the agency's claim at $\alpha = 0.01$?

Soln given, $n_1 = 13$, $S_1 = 27.5$
$n_2 = 16$, $S_2 = 29.75$

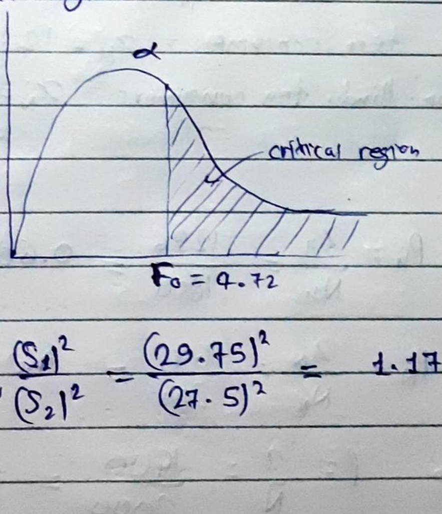
(i) $H_0 : \sigma_1^2 = \sigma_2^2$
$H_1 : \sigma_1^2 \neq \sigma_2^2$ (two tailed, use table for $\alpha/2 \Rightarrow 0.005$)

(ii) ~~$\alpha = 0.00$~~ $\alpha = 0.1$

(iii) $df_N = n_2 - 1 = 16 - 1 = 15$
$df_D = n_1 - 1 = 13 - 1 = 12$

(iv) critical value:-
for $df_N = 15$, $df_D = 12$ & $\alpha = 0.005$ from table
$F_0 = 4.72$

(v) critical region:-



(vi) $F_{statistics} = \frac{(S_1)^2}{(S_2)^2} = \frac{(29.75)^2}{(27.5)^2} = 1.17$

~~The F-statistic lies in critical region, ie we reject H_0 (null hypothesis) (that standard deviations are same) ie We can reject &~~

The F-statistic lies out side of critical region.
ie we ~~reject~~ fail to reject H_0
(that standard deviation of companies are same)
ie Companies claim is reasonable
& we can't reject the claim Ans

Q B4 In a certain sample of 2000 families, 1400 people are consumers of tea. Out of 1800 Hindu families, 1236 families are consumers of tea. State whether there is any significant difference between consumption of tea among Hindu and non-hindu families.

Sol $\Rightarrow$ given, total families $N = 2000$

tea consumers $x = 1400$

Hindu families $= N_H = 1800$

non Hindu familie $= N_N = 200$

Hindu tea consumers $= x_H = 1236$

non-Hindu tea consumers $= x_N = 1400 - 1236 = 164$

Proportions $\Rightarrow$

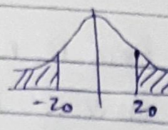
$$P_H = \frac{x_H}{N_H} = \frac{1236}{1800} = 0.6867$$

$$P_N = \frac{x_N}{N_N} = \frac{164}{200} = 0.82$$

Overall proportion $P = \frac{x}{N} = \frac{1400}{2000} = 0.7$

(i) $H_0 : P_H = P_N$

$H_1 : P_H \neq P_N$ — two tailed —



(ii) Critical value for $\alpha = 0.05$ & two tailed (from table)

$$Z_0 = \pm 1.96$$

(iii) Z-statistics for ~~two paired~~ two proportion z-test →

$$Z = \frac{P_H - P_N}{\sqrt{P(1-P)\left(\frac{1}{N_H} + \frac{1}{N_N}\right)}}$$

$$Z = \frac{0.6867 - 0.82}{\sqrt{0.7(1-0.7)\left(\frac{1}{1800} + \frac{1}{200}\right)}}$$

$$\boxed{Z = -3.90}$$

Since Z statistics lie In critical region, we reject H_0

ie

There is a significant difference in tea consumption between hindu and non-hindu families

Q.B-5 Three different kinds of food are tested on three groups of rats for 5 weeks. The objective is to check the difference in mean weight (in grams) of the rats per week. Apply one-way ANOVA using a 0.05 significance level to the following data:

Food I	Food II	Food III
8	4	11
12	5	8
19	4	7
8	6	13
	9	7
6	7	9
11		

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Sol^n H_0: there is no significant difference
H_1: there is significant difference
$C = 3$, (i) $N = 6+6+6 = 18$ (Num of observations)
(ii) T = sum of all values = 154

	x_1	x_2	x_3	Total	x_1^2	x_2^2	x_3^2
	8	4	11	23	64	16	121
	12	5	8	25	144	25	64
	19	4	7	30	361	16	49
	8	6	13	27	64	36	169
	6	9	7	22	36	81	49
	11	7	9	27	121	49	81
Σ	64	35	55	154	794	223	533

(iii) Correction factor → $CF = \frac{T^2}{N} = \frac{(154)^2}{18} = 1317.55$

(iv) Total sum of Squares $TSS = (\Sigma x_1^2 + \Sigma x_2^2 + \ldots) - CF$

$TSS = 794 + 223 + 533 - 1317.55$

$TSS = 232.45$

(v) Column sum of squares =

$SSC = \left(\frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \ldots\right) - CF$

$SSC = \left(\frac{(64)^2}{6} + \frac{(35)^2}{6} + \frac{(55)^2}{6}\right) - 1317.5$

$SSC = 1391 - 1317.5$

$SSC = 73.5$

$SSE = TSS - SSC = 154.95$

(vi) table →

Source of variation	Sum of squares	Degree of freedom	Mean Square	F ratio
Between Columns	SSC = 73.5	C−1 = 2	$MSC = \frac{SSC}{C-1} = \frac{73.5}{2} = 36.75$	(as MSC > MSE) ⇒ $F_c = \frac{36.75}{10.33}$
ERROR	SSE = 154.95	N−C = 15	$MSE = \frac{SSE}{N-C} = \frac{154.95}{15} = 10.33$	$\boxed{F_c = 3.55}$

(vii) critical value :

from table, for $\alpha = 0.05$, $\xrightarrow{C-1}$, $\downarrow N-C$

$F_o = 3.68$

Since $F_{calculated}$ lie in outside of critical region, ie we fail to reject H_o,

Hence there is No Significant difference in mean weight Ans

Q.C1 Certain pesticides are packed into bags by a machine. A random sample of 10 bags is drawn and their contents are found to weight in kgs as follows : 50, 49, 44, 52, 45, 48, 46, 45, 49, 45 Test if the average packing can be taken as 50 kg.

Soln let $\alpha = 0.05$ (standard value)

$$\bar{x} = (50+49+44+52+45+48+46+45+49+45)/10$$

$$\boxed{\bar{x} = 47.3}$$

$$S^2 = \frac{1}{10-1}\sum_{i=1}^{10}(x_i-\bar{x})^2$$

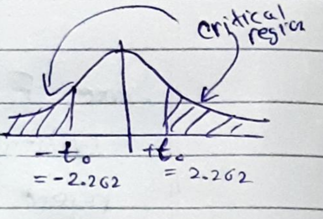
$$S^2 = \frac{1}{9}\Big[(50-47.3)^2+(49-47.3)^2+(44-47.3)^2+(52-47.3)^2+(45-47.3)^2+(48-47.3)^2+(46-47.3)^2+(45-47.3)^2+(49-47.3)^2+(45-47.3)^2\Big]$$

$$S^2 = 7.12$$

$$\boxed{S = 2.66}$$

(i) $H_0 : \mu = 50$

$H_1 : \mu \neq 50$ (two tailed exp)



(ii) $\alpha = 0.05$

(iii) $df = n-1 = 10-1 = 9$

(iv) Critical value →

for $df = 9$, $\alpha = 0.05$, two tailed exp.

$t_0 = \pm 2.262$

(v) test statistic:-

$$t = \frac{\bar{x}-\mu}{\frac{S}{\sqrt{n}}}$$

$$t = \frac{47.3-50}{\frac{2.66}{\sqrt{10}}}$$

$$\boxed{t = -12.14}$$

Since test statistics lie in critical region,
ie we reject null hypothesis.
hence,
The average packing can't be taken as 50kg
Ans

Q. C2 In a test given to two groups of students, the marks obtained are as follows.

First group	18	20	36	50	49	36	34	49	41
Second group	29	28	26	35	30	44	46		

Test if avg marks secured by two groups of students differ significantly.

Sol $n_1 = 9$, $n_2 = 7$

$\bar{x} = (18+20+36+50+49+36+34+49+41)/9 = 37$

$$S_1^2 = \left[(18-37.0)^2 + (20-37.0)^2 + (36-37.0)^2 + (50-37.0)^2 + (49-37.0)^2 + (36-37.0)^2 + (49-37.0)^2 + (41-37.0)^2 + (34-37)^2\right]\frac{1}{8}$$

$S_1 = 11.90$

$\bar{y} = (29+28+26+35+30+44+46)/7 = 34$

$$S_2^2 = \left[(29-34)^2 + (28-34)^2 + (26-34)^2 + (35-34)^2 + (30-34)^2 + (44-34)^2 + (46-34)^2\right]\frac{1}{6}$$

$S_2^2 = 64.33$

$S_2 = 8.02$

(i) $H_0 : \mu_1 = \mu_2$
$H_1 : \mu_1 \neq \mu_2$ (two tailed test)

(ii) $\alpha = 0.05$ (standard value

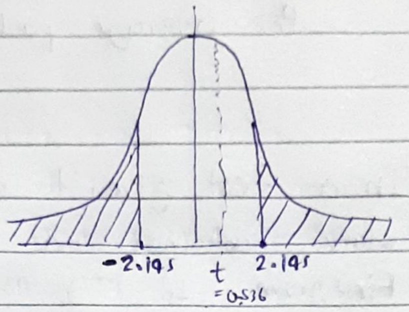
(iii) $df = n_1 + n_2 - 2$
$= 9+7-2 = 14$

(iv) Critical value

for $\alpha = 0.05$ & $dof = 14$, from table

$t_0 = \pm 2.145$

(v) Critical region



(vi) test statistics:

$$t = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{--- (1)}$$

where $S = \sqrt{\dfrac{n_1 S_1^2 + n_2 S_2^2}{(n_1 + n_2 - 2)}}$

$S = \sqrt{\dfrac{9(11.90)^2 + 7(8.02)^2}{(9 + 7 - 2)}}$

$S = 11.09$

hence test statistics:-

$$t = \frac{(37 - 34) - (0)}{11.09\sqrt{\frac{1}{9} + \frac{1}{7}}}$$

$t = 0.536$

Since t lie outside of critical region,
ie we can't reject H_0 (null hypothesis)
hence avg marks secured by two groups
groups of people do not differe significantly

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Q C3 Two random samples were drawn from two normal populations and their values are as follows:

A	66	67	75	76	82	84	88	90	92		
B	64	66	74	78	82	85	87	92	93	95	97

Test whether the two populations have the same variance at 10% level of significance

Soln

$\bar{x}_A = (66+67+75+76+82+84+88+90+92)/9 = 80$

$S_A^2 = \left[(66-80)^2+(67-80)^2+(75-80)^2+(76-80)^2+(82-80)^2+(84-80)^2 + (88-80)^2+(90-80)^2+(92-80)^2\right]\frac{1}{8}$

$\boxed{S_A = 9.57}$

$\bar{x}_B = (64+66+74+78+82+85+87+92+93+95+97)/11$

$\bar{x}_B = 83$

$S_B^2 = \left((64-83)^2+(66-83)^2+(74-83)^2+(78-83)^2+(82-83)^2+(85-83)^2 + (87-83)^2+(92-83)^2+(93-83)^2+(95-83)^2+(97-83)^2\right)\frac{1}{10}$

$\boxed{S_B = 11.39}$

(F test for variances) ⇒

(i) $H_0: \sigma_A^2 = \sigma_B^2$

$H_1: \sigma_A^2 \neq \sigma_B^2$ (two tailed)

(ii) $F_{statistic} = \frac{S_B^2 \text{ (larger)}}{S_A^2 \text{ (smaller)}} = 1.416$

(iii) $df_N = n_2 - 1 = 10$ (given $n_1 = 9$, $n_2 = 11$)

$df_D = n_1 - 1 = 8$

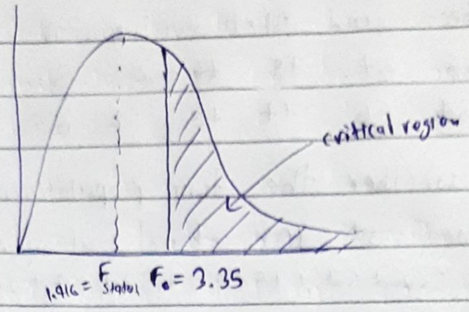
(iv) critical value →

for $df_N = 10$, $df_D = 8$ from $\frac{\alpha}{2} = \frac{0.1}{2} = 0.05$ table

as (two tailed test)

$F_0 = 3.35$

(v) critical region –



Since F-statistic lie out of critical region, ie we fail to reject the null hypothesis.
ie two population have same variance at 10% level of significance Ans

Q C4 1000 families were selected at random in a city to test the belief that the high income group families usually send their children to public schools and the low income families often send their children to government schools. The following results were obtained

Income	Public School	Govt. School	Total
Low	370	430	800
High	130	70	200
Total	500	500	1000

Test whether income and type of schooling are independent

Sol $\because$ expected frequency ≥ 5 & observations are randomly selected, the chi-square test can be used.

(i) H_0: income and type of schooling are independent

H_1: income and type of schooling are dependent

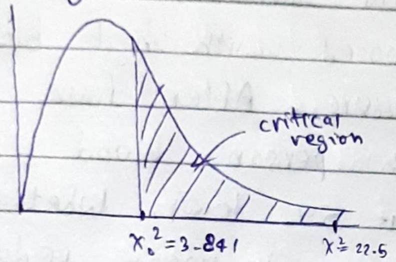
(ii) $\alpha = 0.05$ (standard value)

(iii) $df = (r-1)(c-1) = (2-1)(2-1) = 1$

(iv) Critical value:

$$\chi_0^2 = 3.841 \quad \text{(from it's table for } df=1,\ \alpha=0.05)$$

(v) Critical region:



(vi) Expected frequency

$$E_{r,c} = \frac{(\text{Sum of row } r) \times (\text{sum of Column } c)}{\text{Sample size}}$$

$$E_{1,1} \quad \frac{(500)(800)}{1000} = 400$$

$$E_{1,2} = \frac{(800)(500)}{1000} = 400$$

$$E_{2,1} = \frac{(200)(500)}{1000} = 100$$

$$E_{2,2} = \frac{(200)(500)}{1000} = 100$$

(vii) $$\chi^2 = \Sigma \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

$$= \frac{(370-400)^2}{400} + \frac{(430-400)^2}{400} + \frac{(130-100)^2}{100} + \frac{(70-100)^2}{100}$$

$$\chi^2 = 22.5$$

Since χ^2 lie in critical region, ie we reject Null hypothesis (H_0), so By that,
Income and type of schooling are dependent <u>Ans</u>

Q C5 Three different techniques, namely medication, exercies and special diet are randomly assigned to (individuals diagnosed with high blood pressure) lower blood pressure. After four weeks the reduction in each person's blood pressure is recorded. Test at 5% level, whether there is significant difference in mean reduction of blood pressure among the three techniques.

Medication	10	12	9	15	13
Exercise	6	8	3	0	2
Diet	5	9	12	8	4

ANOVA

Sol→ H_0 : $\mu_1 = \mu_2 = \mu_3$ (No significant differences)
H_1 : there is significant diff among means

$c = 3$

(i) $N = 5+5+5 = 15$ (num of observations)

(ii) $T =$ sum of all values $= 116$

x_1	x_2	x_3	Total	x_1^2	x_2^2	x_3^2
10	6	5	21	100	36	25
12	8	9	29	144	64	81
9	3	12	24	81	9	144
15	0	8	23	225	0	64
13	2	4	19	169	16	16
Σ 59	19	38	116	719	125	330

(iii) Correction factor = $CF = \frac{T^2}{N} = \frac{(116)^2}{15} = 897.06$

(iv) Total sum of Squares ⇒ $TSS = (\Sigma x_1^2 + \Sigma x_2^2 + \ldots) - CF$

$TSS = (719 + 125 + 330) - 897.06$

$TSS = 276.94$

(v) Column sum of square ⇒

$$SSC = \left(\frac{(\Sigma x_1)^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \ldots\right) - CF$$

$$SSC = \left(\frac{(59)^2}{5} + \frac{(19)^2}{5} + \frac{(38)^2}{5}\right) - 897.06$$

$SSC = 160.14$

$SSE = TSS - SSC = 276.94 - 160.14 = 116.8$

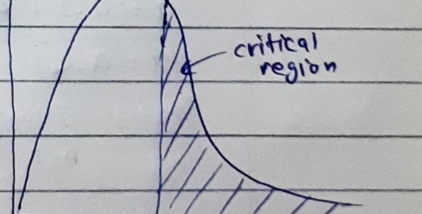
(vi) table ⇒

Source of Variation	Sum of Squares	Degree of freedom	Mean Square	F ratio
Between Columns	$SSC = 160.14$	$C-1 = 3-1 = 2$	$MSC = \frac{SSC}{C-1} = \frac{160.14}{2} = 80.07$	(as MSC > MSE) $F_c = \frac{80.07}{9.73}$
ERROR	$SSE = 116.8$	$N-C = 15-3 = 12$	$MSE = \frac{SSE}{N-C} = \frac{116.8}{12} = 9.73$	$\boxed{F_c = 8.22}$

(vii) Critical value ⇒

from table for $\alpha = 0.05$, $C-1 = 2$, $N-C = 12$

$F_0 = 3.89$



Since $F_{calculated}$ lie in Critical Region, ie We Reject H_0,
hence, There is significant difference in mean reduction of blood pressure among the three techniques. Ans

Q.21 The lifetime of electric bulbs for a random sample of 10 from a large consignment gave the following data: Life in thousand hours: 4.2, 4.6, 4.1, 3.9, 5.2, 3.8, 3.9, 4.3, 4.4, 5.6. Can we accept the hypothesis that the average lifetime of bulbs is 4000 hours?

Soln (i) $H_0 : \mu = 4000$ (in thousand)

$H_1 : \mu \neq 4000$ (in thousand) (two tailed t-test)

(ii) $\alpha = 0.05$ (standard value)

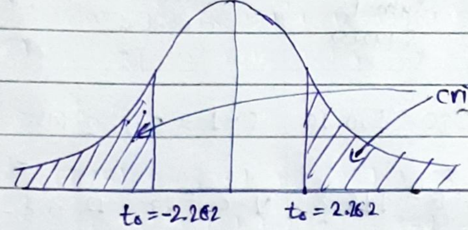
(iii) degree of freedom $df = n-1 = 10-1 = 9$

(iv) critical values ⇒

by table for $\alpha = 0.05$, $df = 9$

$$t_0 = \pm 2.262$$

(v) critical region ⇒



(vi) test statistics ⇒

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}} \quad \text{——} ①$$

$$\bar{x} = (4.2 + 4.6 + 4.1 + 3.9 + 5.2 + 3.8 + 3.9 + 4.3 + 4.4 + 5.6)\frac{1}{10}$$

$$\boxed{\bar{x} = 4.4}$$

$$S^2 = \frac{1}{n-1} \sum_{1}^{n} (x_i - \bar{x}) = \frac{1}{9} \sum_{1}^{10} (x_i - 4.4)$$

$$S^2 = 0.3466$$

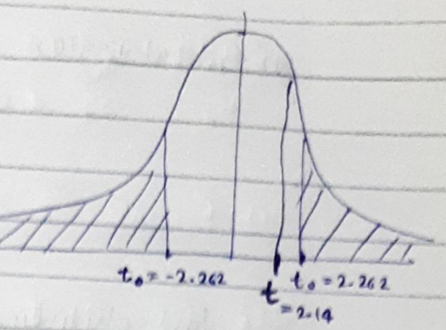
$$\boxed{S = 0.588}$$

$$\boxed{n = 10}$$

put up all values in equation ①

$$t = \frac{4.4 - 4.0}{\frac{0.5887}{\sqrt{10}}}$$

$$\boxed{t = 2.14}$$



Since test statistic lie in non-critical region, ie we accept null hypothesis (H_0)
hence
We can accept the hypothesis that the average lifetime of bulbs is 4000 hours.
Ans

Q D2 For a random sample of 10 persons fed on died A the increased weight in pounds in a certain period were 10, 6, 16, 17, 13, 12, 8, 14, 15 and 9. For another random sample of 12 persons fed on died B, the increase in the same period ~~a~~ were 7, 13, 22, 15, 12, 14, 18, 8, 21, 23, 10 and 17. Test whether the diets A and B differ significantly as regards their effect on increase on weight.

Sol → t-test → (given $n_1 = 10, n_2 = 12$)

(i) $H_0 : \mu_A = \mu_B$

$H_1 : \mu_A \neq \mu_B$ (two tailed exp)

(ii) $\alpha = 0.05$ (standard value)

(iii) $df = n_1 + n_2 + 2$

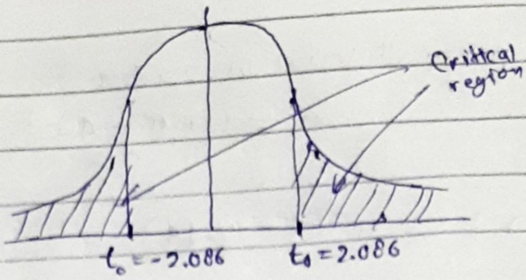
$= 10 + 12 - 2 = 20$

(iv) Critical value →

for $\alpha = 0.05$ & dof $= 20$ from table

$t_0 = \pm 2.086$

(v) Critical region -



(vi) test statistics ⇒

$$t = \frac{(\bar{x}-\bar{y}) - (\mu_A - \mu_B)}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{—— ①}$$

where $$S = \sqrt{\frac{n_1 S_A^2 + n_2 S_B^2}{(n_1 + n_2 - 2)}} \quad \text{—— ②}$$

* $\bar{x} = (10+6+16+17+13+12+8+14+15+9)/10 = \boxed{12}$
* $\bar{y} = (7+13+22+15+12+14+18+8+21+23+10+17)/12 = \boxed{15}$
* $S_A^2 = \frac{1}{9}\left((10-12)^2 + (6-12)^2 + (16-12)^2 + (17-12)^2 + (13-12)^2 + (12-12)^2 + (8-12)^2 + (14-12)^2 + (15-12)^2 + (9-12)^2\right)$

$\boxed{S_A^2 = 13.33}$

* $S_B^2 = \frac{1}{11}\left((7-15)^2 + (13-15)^2 + (22-15)^2 + (15-15)^2 + (12-15)^2 + (14-15)^2 + (18-15)^2 + (8-15)^2 + (21-15)^2 + (23-15)^2 + (10-15)^2 + (17-15)^2\right)$

$\boxed{S_B^2 = 28.54}$

putting up all values

$$S = \sqrt{\frac{(10)(13.33) + (12)(28.54)}{(10+12-2)}}$$

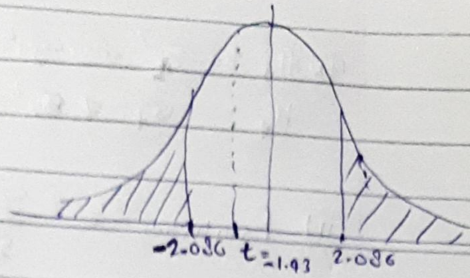
$$S = 4.877$$

So test statistics

$$t = \frac{(\bar{x}-\bar{y}) - (\mu_A - \mu_B)}{S\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$t = \frac{(12-15)^{*} - (0)}{4.877\sqrt{\frac{1}{10} + \frac{1}{12}}}$$

$$\boxed{t = -1.43}$$



Since test statistics lie in non critical region ie we fail to reject H_0 hence
There is no significant difference between diets A & B regards their effect on weight increase. Ans

Q3 Consider the following measurement of the heat producing capacity of the coal produced by two mines (in millions of calories per ton):

Mine 1	8260	8130	8350	8070	8340	
Mine 2	7950	7890	7900	8140	7920	7840

Can it be concluded that the two population variances are equal?

Soln (F-test for variances) →

$\bar{x}_1 = (8260 + 8190 + 8350 + 8070 + 8340)/5 = \underline{8230}$

$$S_1^2 = \frac{1}{4}\left((8260-8230)^2 + (8130-8230)^2 + (8350-8230)^2 + (8070-8230)^2 + (8340-8230)^2\right)$$

$S_1^2 = \underline{15750}$

$\bar{x}_2 = (7950 + 7890 + 7900 + 8140 + 7920 + 7840)/6 = 7940$

$$S_2^2 = \frac{1}{5}\left((7950-7940)^2 + (7890-7940)^2 + (7900-7940)^2 + (8140-7940)^2 + (7920-7940)^2 + (7840-7940)^2\right)$$

$S_2^2 = \underline{10920}$

(i) $H_0 : \sigma_1^2 = \sigma_2^2$

$H_1 : \sigma_1^2 \neq \sigma_2^2$ (two tailed)

(ii) $F_{statistics} = \dfrac{S_1^2 \text{ (larger)}}{S_2^2 \text{ (smaller)}} = \dfrac{15750}{10920} = 1.442$

(iii) $df_N = n_1 - 1 = 5 - 1 = 4$

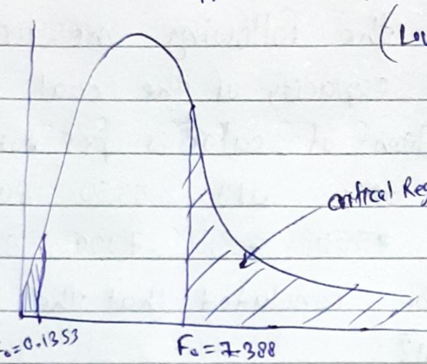
$df_D = n_2 - 1 = 6 - 1 = 5$

(iv) Critical values →

for $df_N = 4$, $df_D = 5$ from $\frac{\alpha}{2} = \frac{0.05}{2} = 0.025$ table

$F_0 = 7.388$ (upper critical value)

(Lower critical value $= \frac{1}{7.388} = 0.1353$



Since Fstatistics lie out side of critical Region

ie we can not reject Null hypothesis.

ie we do not have enough evidence to say that the variences are not equal.

Q. 04 Three composition instructions recorded the number of spelling errors which their students made on a research paper. At 1% level of significance test whether there is significant difference in the in average number of errors in the three classes of students.

Instructor 1	2	3	5	0	8		
Instructor 2	4	6	8	4	9	0	2
Instructor 3	5	2	3	2	3	3	

ANOVA

Sol ⇒

H_0: there is no significant difference ($\mu_1 = \mu_2 = \mu_3$)

H_1: there is significant difference among means.

$C = 3$

(i) $N = 5 + 7 + 6 = 18$ (num of observations)

(ii) $T =$ sum of all values $= 69$

	X_1	X_2	X_3	Total	X_1^2	X_2^2	X_3^2
	2	4	5	11	4	16	25
	3	6	2	11	9	36	4
	5	8	3	16	25	64	9
	0	4	2	6	0	16	4
	8	9	3	20	64	81	9
		0	3	3		0	9
		2		2		4	
Σ	18	33	18	69	102	217	60

(iii) correction factor $= CF = \frac{T^2}{N} = \frac{(69)^2}{18} = 264.5$

(iv) Total sum of Squares ⇒

$TSS = (\Sigma x_1^2 + \Sigma x_2^2 + \cdots) - CF$

$TSS = (102 + 217 + 60) - 264.5$

$TSS = 114.5$

(v) Column sum of squares ⇒

$$SSC = \left(\frac{(\Sigma (x_1))^2}{N_1} + \frac{(\Sigma x_2)^2}{N_2} + \cdots\right) - CF$$

$$SSC = \left(\frac{(18)^2}{5} + \frac{(33)^2}{7} + \frac{(18)^2}{6}\right) - 264.5$$

$$SSC = 9.87$$

$$SSE = TSS - SSC = 114.5 - 9.87 = 104.63$$

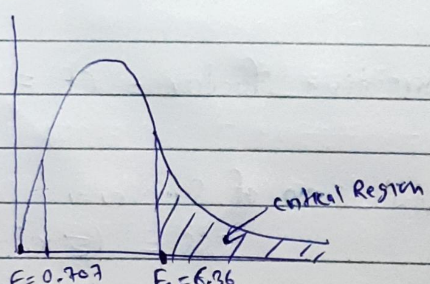
(vi) table ⇒

Source of variation	Sum of squares	Degree of freedom	Mean Square	F-ratio
Between columns	$SSC = 9.87$	$c-1 = 3-1 = 2$	$MSC = \frac{SSC}{c-1} = \frac{9.87}{2}$ $MSC = 4.935$	
ERROR	$SSE = 104.63$	$N-c = 18-3 = 15$	$MSE = \frac{SSE}{N-c} = \frac{104.63}{15}$ $MSE = 6.975$	$F = \frac{MSC}{MSE} = \frac{4.935}{6.975}$ $\boxed{F = 0.707}$

(vii) Critical value ⇒

from table for $\alpha = 0.01$, $F_N = 2, F_D = 15$

$\boxed{F_0 = 6.36}$



Since F statistic lie in non critical region we can't reject null hypothesis (H_0) that there is no significant diff. in means.